

ENGINE AUDIT REPORT — HOME MAINTENANCE

StreamNavigator — Engine Audit
2026-09-20

An evaluation of `/home-maintenance` — the page, the intake, the engine behind it, and whether a customer who pays \$59 gets more than \$59 back.

Conducted 2026-09-20 against `home-maintenance.html`, `api/\_lib/navigator-engine.js`, `prices.config.json` and `tests/` at the repository's current `main`. No paid model runs were spent — the API budget is closed (see `docs/INSURANCE-AUDIT.md`'s Executive Summary and `[[no-api-credit-topups]]`), so this rests on the live page and the engine's own source. Found as a direct consequence of auditing `/insurance`: checking every other product for the same defect class turned up two more, of which this is the first fixed.

Executive Summary

Home Maintenance Navigator sold a repair-vs-replace verdict "grounded in real numbers," and delivered a single model call told to reason from "general knowledge of typical lifespans and typical cost ranges."

`api/\_lib/navigator-engine.js`'s `PRODUCT\_CONFIGS['home-maintenance']` task, verbatim: *"Weigh expected remaining life if repaired vs. replaced using general knowledge of typical lifespans and typical cost ranges for that system and repair type."* That is the identical unverifiable claim `docs/INSURANCE-AUDIT.md` found and removed from Insurance Navigator, and the identical one this codebase's earlier audits had already banned from Home Savings and Government Money — this codebase holds no price table for what a roof, an HVAC system, or a water heater "usually" costs to repair or replace, and repair/replacement cost depends on the specific unit, the contractor and the region far more than any general-knowledge guess could responsibly account for.

Two structural defects compounded it. First, the intake was a single free-text box plus an optional file upload — nothing forced a customer to supply the one input a real financial comparison needs (their own repair and replacement figures), so a customer could pay \$59 having typed a single sentence and received exactly the ungrounded guess the page implicitly promised not to be. Second, nothing on the page or in the prompt treated a safety hazard differently from ordinary wear — a gas smell or a sparking outlet was just more input for the same cost-weighting prompt, when the honest answer to "is this worth repairing" is "stop and call someone now," not a dollar comparison.

Both are fixed. `navigator-home-maintenance-engine.js` is a deterministic decision engine: it never compares to a "typical" cost, it will not render a financial verdict without the customer's own repair and/or replacement quotes, and a flagged safety concern short-circuits straight to urgent guidance before any cost math runs, unconditionally. The intake now collects structured age, symptom and quote fields instead of free text, gated server-side the same way Buying, Home Savings, Government Money, Landlord and Insurance already are.

One deliberate, disclosed exception exists in the new engine, and it is worth stating plainly rather than leaving implicit: a small curated table of typical service-LIFE ranges, in years, by system category. This is not a relapse into the banned pattern. A current repair or replacement COST is a market price with no stable source; a system's typical SERVICE LIFE is a slow-moving engineering fact, commonly published by manufacturers and standards bodies, that does not reprice itself by region or quarter. It is used only as context — where does this system's age sit on a normal range — never to produce or imply a dollar figure, and every statement of it carries its own caveat about material and regional variation. The file header

documents the distinction explicitly so it can be checked, not just trusted.

Promise vs. Delivery

Promise	What the Product Delivered (Before)	Gap	Severity	Fix
"Get a clear repair-vs-replace recommendation grounded in real numbers"	A model call reasoning from its own general knowledge of typical costs — not the customer's numbers, because the intake never required any	The word "real" described the opposite of what the prompt instructed	Critical	<code>`navigator-home-maintenance-engine.js`</code> : verdict is arithmetic on the customer's own quotes, or an explicit request for what's missing — never a guess
"Typical cost ranges for this system and repair/replacement type" (What You Get)	The exact banned claim, asserted as a feature	This is the claim <code>`docs/INSURANCE-AUDIT.md`</code> , <code>`HOME-SAVINGS-AUDIT.md`</code> and <code>`GOVERNMENT-MONEY-AUDIT.md`</code> each independently found and removed elsewhere	Critical	Bullet removed from the page; the prompt now forbids stating any cost that is not the customer's own quote
"Efficiency and cost-of-ownership differences"	Nothing computed this; the model was simply asked to mention it	A promise with no mechanism behind it at all	High	Removed from the page rather than built on a guess — see Required Changes
"Warning signs it's time to stop patching and replace"	Generic, ungrounded prose	Plausible but not tied to anything the customer actually told the product	Medium	Now driven by the deterministic <code>`cautions`</code> array (e.g. a system already past its typical service-life range)
Intake: a free-text box, files optional	A customer could submit a single sentence and reach checkout	No structured gate meant no guaranteed inputs for the "real numbers" the page promised	Critical	Structured fields (age, symptoms, both quotes) with a server-side sufficiency gate matching the pattern already used on five other products

Promise	What the Product Delivered (Before)	Gap	Severity	Fix
(Implicit) a safety hazard is handled like ordinary wear	The same cost-weighting prompt ran regardless of what prompted the decision	A gas smell treated as a data point in a cost comparison is the one outcome this kind of product must never produce	Critical	<code>`applySafetyOverride()`</code> — an unconditional short-circuit, tested directly, that cost math can never reach if <code>`safety_concern`</code> is flagged

Fixes Shipped

Critical

1. Deterministic decision engine (``navigator-home-maintenance-engine.js``) — no "typical cost" claim anywhere in its logic or its output.
1. Structured intake replacing the free-text box, with a server-side sufficiency gate (``api/navigator-intake.js``) matching Buying/Home Savings/Government Money/Landlord/Insurance.
1. An unconditional safety override that cost comparison can never reach.

High

1. "Efficiency and cost-of-ownership differences" removed from the page rather than built on an ungrounded estimate — the audit's own standard (``docs/INSURANCE-AUDIT.md`` §13, "identify bad features too") applied here directly: a feature with no honest mechanism behind it should be cut, not faked.
1. The write-up prompt (``PRODUCT_CONFIGS['home-maintenance'].task``) now forbids originating any dollar figure, any lifespan claim beyond the engine's own caveated range, or any softening of a safety verdict.

Medium

1. Hero example-finding card added, proving the mechanism (the FAQ's own original placeholder scenario — a 14-year-old water heater, \$1,400 repair vs. \$2,600 replace) before a visitor pays.
1. FAQ and footer disclaimer corrected to describe what the product actually does — that a quote is not strictly required to submit, but is required for a financial verdict, and that no cost data is held or estimated.

20 new tests (``tests/home-maintenance-engine.test.js``, ``tests/navigator-intake-home-maintenance.test.js``) pin every verdict path — safety override, both-quotes comparison at and around the 50% threshold, each single-missing-quote case, the no-baseline case, and the sufficiency gate — against known answers. All 83 suites in the repository pass.

Pricing

Current price: \$59, one-time. No change recommended, on the same terms ``docs/INSURANCE-AUDIT.md`` set for Insurance and this codebase already applied to Landlord and Government Money: the price stands once its conditions are met, and they now are — a real deterministic engine exists, the "typical cost" claim is gone, and the intake requires the inputs the verdict actually needs. ``prices.config.json`` carries a note recording this.

What Remains Unverifiable

The same shape as Insurance's remaining gap, and smaller: this engine needs no model call to reach a verdict at all — ``analyze()`` is pure arithmetic on structured form fields, fully covered by ``tests/home-maintenance-engine.test.js`` without spending anything. The only unverified piece is the WRITE-UP: does the model, handed the computed verdict as data, present it without inventing a figure or softening a safety verdict, in a real generation? That is downstream of the same blocked Anthropic credit balance ``docs/INSURANCE-AUDIT-FOLLOWUP.md`` documents, not something more engine work can close.